mmWave UE Positioning Under Blockage: Limits for ISAC-Assisted Handover

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***Abstract*—Blockage-aware proactive handover in millimeter-wave (mmWave) networks switches a user to another cell before a body crosses the line of sight (LoS) to its radio unit. Predicting such a crossing requires knowing where the user is, so the scheme is only as good as the network's positioning. This paper asks whether the radio units that serve a user can localize it accurately enough from its uplink sounding reference signals. We derive single-epoch position error bounds that treat synchronization and array calibration errors as Bayesian priors, compare LoS-only information with map-aided information from a digital twin, and evaluate a dominant-path estimator and a map-aided multipath tracker. Ray-traced experiments in a street canyon, reported on held-out scenarios, show that array calibration, not bandwidth, limits the bound, and that blockage substantially degrades the LoS-only bound, whereas known reflected paths keep the map-aided bound much lower. The dominant-path estimator has mostly systematic errors that grow sharply around blockage events, and its positions make proactive handover worse than 3GPP event-triggered handover. The map-aided tracker reduces the median error from 12 to 3.7 cm and keeps it low around blockage onsets; with its positions, proactive handover is no longer significantly worse than the 3GPP event, but not better either.**

***Index Terms*—mmWave positioning, position error bound, uplink SRS, blockage, proactive handover, digital twin, ISAC.**

I. Introduction

Millimeter-wave (mmWave) bands provide the bandwidth that 5G and 6G networks need for high data rates, but their links are fragile: a pedestrian or vehicle crossing the line of sight (LoS) attenuates the signal by tens of decibels within milliseconds [1], [2]. Reactive handover cannot avoid such outages. Integrated sensing and communication (ISAC) and other sensing-aided approaches therefore aim to predict blockage and hand the user over in advance [3]–[5].

Predicting a blockage means predicting whether a body will cross the line between a radio unit and the user. The prediction can thus be no better than the network's knowledge of where the user is. In a companion study [6], we found that a blockage-aware handover planner outperforms the service-aware 3GPP A5 event only if the user position is known with an error of about 8–13 cm (per horizontal axis), and that a meter-level error removes the benefit entirely.

Can the network meet this requirement with the infrastructure it already has? 5G NR supports uplink positioning from sounding reference signals (SRS) [7], and recent street-scale studies with several base stations report errors below a meter [8]. Position error bounds for mmWave systems suggest that much better accuracy is possible in principle [9], [10], but they typically consider ideal hardware or a single impairment in static scenes [11]. It remains unclear which effect limits decimeter accuracy in practice, how blockage changes it, how close practical estimators come, and whether their errors are small enough at the moments when a handover decision is made.

This paper addresses these questions in a street canyon served by two O-RAN radio units (O-RUs). Our contributions are:

- A positioning analysis tailored to blockage-aware handover: a single-epoch equivalent Fisher information analysis with synchronization and array calibration errors as Bayesian priors, two models of a blocked LoS, and map-aided information from reflected paths with known geometry.
- Two practical uplink estimators: a dominant-path estimator with an extended Kalman filter (EKF), whose single-epoch front end is compared with the bound by decomposing its error into bias and random spread, and a map-aided multipath tracker that associates extracted paths with virtual anchors of the digital twin.
- An analysis of positioning errors around blockage onsets and a closing experiment that feeds estimated and synthetic bound-level positions into a blockage-aware handover planner. The bounds, the estimator, and the scripts that generate every number in this paper are released as open source.[1]

II. Related Work

Bounds for mmWave positioning. Fisher information analyses show that the angle and delay resolution of mmWave arrays support sub-decimeter accuracy [9], [10], building on the equivalent Fisher information framework for wideband localization [12]. Multipath components with known geometry carry position information as virtual anchors [13], [14], and multipath tracking associates them with such anchors probabilistically [15]. We add hardware errors as priors and evaluate bounds and estimators under traffic-driven blockage.

Uplink SRS positioning. 3GPP specifies uplink time- and angle-based positioning [7]. Dong *et al.* [8] combine

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[1]https://github.com/CLIS-WPI/sense2connect

LoS/NLoS gating, dominant-path angle and delay estimation, and multi-base-station EKF fusion, reach sub-meter errors in a ray-traced campus, and point to reflections as virtual anchors for future work.

Sensing-aided handover. Vision-, radar-, and ISAC-aided handover predicts blockage or user motion [4], [5]; our companion study [6] identified the user position as a binding requirement.

III. System Model and Problem Formulation

A. Signal Model

A single-antenna user equipment (UE) at horizontal position $\mathbf{p}$ at known height (distances and directions are evaluated at the 3D point $[\mathbf{p}^\mathsf{T}, z_{\mathrm{UE}}]^\mathsf{T}$) transmits an uplink SRS that is received by a set $\mathcal{C}$ of O-RUs, each with an M-element array. At O-RU c, subcarrier frequency f_n, and element m, the received sample is

$$y_{c,n,m} = \sum_{\ell} \beta_{c,\ell}\, e^{-j2\pi f_n(\tau_{c,\ell}+\delta_c+b)}\, e^{j\psi_{c,m}}\, s_m(\mathbf{u}_{c,\ell}) + w_{c,n,m}, \tag{1}$$

where path ℓ has complex gain $\beta_{c,\ell}$ (including blockage loss), delay $\tau_{c,\ell}$, and arrival direction $\mathbf{u}_{c,\ell}$ with steering response s_m; δ_c is the time offset of O-RU c, b the UE clock bias, $\psi_{c,m}$ the calibration phase error of element m, and $w_{c,n,m} \sim \mathcal{CN}(0, \sigma_w^2)$ independent thermal noise. Hardware errors are random with $\delta_c \sim \mathcal{N}(0, \sigma_{\mathrm{sync}}^2)$ and $\psi_{c,m} \sim \mathcal{N}(0, \sigma_\phi^2)$.

B. Geometry and Blockage

The LoS delay and angles are functions of $\mathbf{p}$ and the O-RU position $\mathbf{p}_c$, e.g., $\tau_{c,0} = \|\mathbf{p} - \mathbf{p}_c\|/c_0$. A specular reflection on a known surface is an LoS path from the virtual anchor $\tilde{\mathbf{p}}_{c,\ell}$, the image of $\mathbf{p}_c$ in that surface, so that $\tau_{c,\ell} = \|\mathbf{p} - \tilde{\mathbf{p}}_{c,\ell}\|/c_0$ [13]. Moving bodies attenuate every path segment according to a blockage model; we use 3GPP model B [16]. An O-RU is *blocked* for a UE when the loss of its LoS path exceeds a threshold. Model B reduces the power of a blocked LoS but keeps its geometry, whereas the field that actually arrives diffracts around the blocker. We therefore consider two models of a blocked LoS: *kept*, in which its delay and angles still follow $\mathbf{p}$ (optimistic), and *biased*, in which they carry an unknown bias. In the biased model, a bound treats them as free parameters, and an estimator observes a blocked LoS whose delay and direction deviate from the geometry of $\mathbf{p}$.

C. Positioning Requirement of Blockage-Aware Handover

A blockage-aware handover policy π predicts the SNR of each cell from the tracked blockers and the reported UE position $\hat{\mathbf{p}}_t = \mathbf{p}_t + \mathbf{e}_t$, and selects the serving cell accordingly. Its outage time $J(\pi; \{\hat{\mathbf{p}}_t\})$ counts the time below the SNR required for the service rate, including the interruptions of its handovers. Since the policy aims to outperform a reactive reference policy π_0 (here, the 3GPP A5 event), the positioning requirement is that the error process $\{\mathbf{e}_t\}$ keeps

$$J\big(\pi; \{\hat{\mathbf{p}}_t\}\big) \le J(\pi_0). \tag{2}$$

This requirement depends on the error at the times the policy acts, not only on its overall statistics. We therefore examine (i) what limits the position error bound, (ii) how blockage and map knowledge change it, (iii) how a practical estimator compares, and (iv) whether its errors satisfy (2).

IV. Position Error Bound and Estimator

A. Single-Epoch Position Error Bound

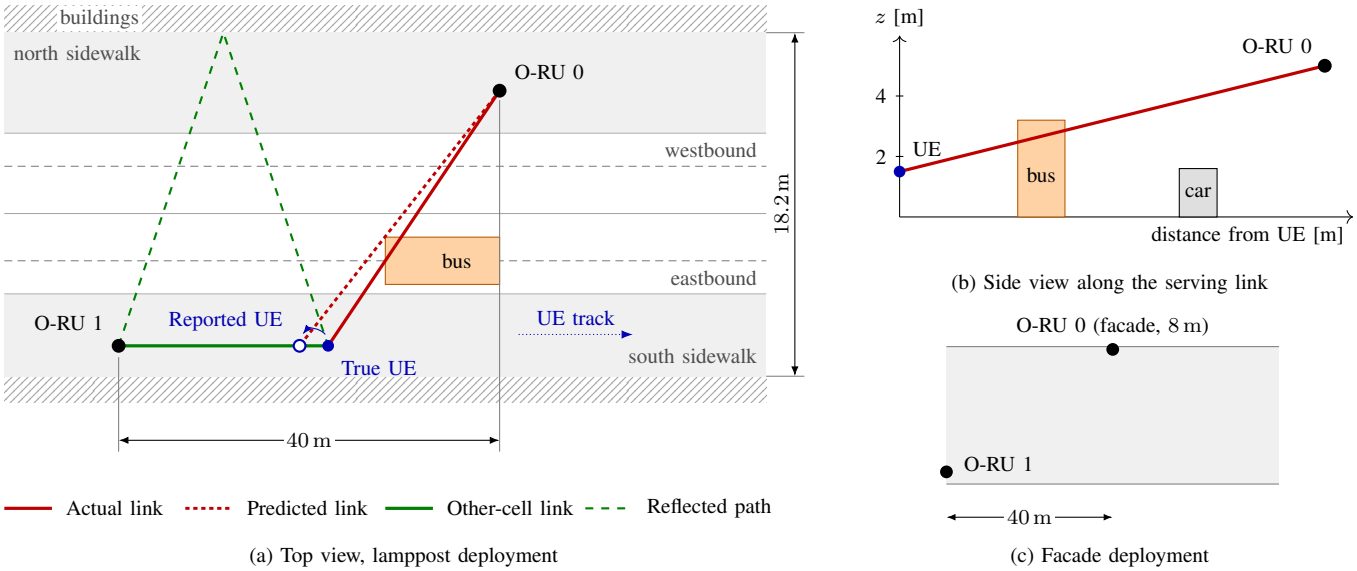
Collect the channel parameters of all O-RUs at one epoch in $\boldsymbol{\eta}$ (delays, angles, complex gains) and the hardware and timing errors in $\boldsymbol{\nu} = [\{\delta_c\}, b, \{\psi_{c,m}\}]$. The Fisher information of $\boldsymbol{\eta}$ follows from (1) by the Slepian–Bangs formula; we compute it exactly by factorizing its frequency and antenna sums. With $\mathbf{T} = \partial\boldsymbol{\eta}/\partial\mathbf{p}$, the equivalent Fisher information of the position is

$$\mathbf{J}_{\mathbf{p}} = \mathbf{T}^\mathsf{T}\mathbf{J}_{\boldsymbol{\eta}}\mathbf{T} - \mathbf{T}^\mathsf{T}\mathbf{J}_{\boldsymbol{\eta}\boldsymbol{\nu}}\big(\mathbf{J}_{\boldsymbol{\nu}\boldsymbol{\nu}} + \boldsymbol{\Lambda}\big)^{-1}\mathbf{J}_{\boldsymbol{\nu}\boldsymbol{\eta}}\mathbf{T}, \tag{3}$$

where all blocks are those remaining after the channel nuisance parameters have been eliminated jointly by a Schur complement: the complex gains of all paths; in the *LoS-only* bound, the delays and angles of all reflected paths; and in the biased model, the delay and angles of a blocked LoS. The second term eliminates the timing and hardware errors in $\boldsymbol{\nu}$. $\boldsymbol{\Lambda}$ holds the Gaussian priors of the O-RU time offsets (σ_{sync}) and of the element phases (σ_ϕ); the UE clock bias has no prior (time difference of arrival, TDoA). In the *map-aided* bound, the delays and angles of the reflected paths are not nuisance parameters: their geometry is known from the digital twin, and their entries of $\mathbf{T}$ follow analytically from the virtual anchors. The analytic derivatives agree with central finite differences at 1000 random UE positions to a relative error of at most 3.2×10^{-6}, and the GPU implementation of (3) matches an independent NumPy implementation to 2.9×10^{-6} and a brute-force numerical Fisher matrix on toy cases to 6.6×10^{-10}; further checks are in the repository. The position error bound is $\mathrm{PEB} = \sqrt{\mathrm{tr}(\mathbf{J}_{\mathbf{p}}^{-1})}$. Because the hardware errors enter through priors, it is a hybrid Cramér–Rao bound: it bounds the root-mean-square (RMS) error of unbiased single-epoch estimates for a given geometry, averaged over the noise and the hardware priors; it is neither a median error nor a guarantee that an estimator attains it, a biased estimator can have a smaller spread, and it excludes the temporal information that a tracking filter accumulates.

B. Dominant-Path Estimator

Per O-RU and epoch, a zero-padded two-dimensional DFT over the array selects the strongest beam; a Hann-windowed delay transform of the beamformed signal gives the delay of the dominant path, refined by parabolic interpolation and followed by an angle refinement at that delay. A peak-to-second-peak test flags reflected dominant paths. Because a planar array of isotropic elements cannot distinguish the two sides of its plane, each angle yields two direction hypotheses; the estimator keeps the joint hypothesis that is consistent across O-RUs, lies in the walkable area of the digital-twin



(a) Top view, lamppost deployment

(b) Side view along the serving link

(c) Facade deployment

Fig. 1. Scenario geometry (dimensions indicated; axes scaled independently). (a) Two neighboring cells share the street; O-RU 0 and O-RU 1 are on 5 m lampposts, 40 m apart along the street. The true serving link (solid red) crosses the bus, whereas the link predicted from the reported UE position (dotted) passes it, so a position error turns into a missed blockage prediction; green: link to the other cell and one of its facade reflections (dashed). (b) The LoS passes through a bus but above a car. (c) In the facade deployment, O-RU 0 is mounted on the north facade at 8 m.

map, and passes the gate of a constant-velocity EKF, which fuses weighted least-squares fixes over time.

C. Map-Aided Multipath Tracker

Per O-RU and epoch, a sparse extraction with continuous refinement returns up to eight multipath components with variances from the local Fisher information. The sources are the LoS and every single- and double-bounce virtual anchor of the street-facing building faces and the ground that is geometrically valid at the predicted position. Loopy belief propagation computes the probabilities that a source generated a component or that the component is clutter, with constant visibility priors and a blocked mode that models a detoured LoS by an inflated covariance [15]; a probabilistic-data-association EKF with a constant-velocity model then updates the position, velocity, UE clock bias, and O-RU time offsets, and projects the position onto the walkable area. A grid search over the walkable area initializes the track and re-initializes it after a loss of association.

V. Experiments and Results

A. Setup

Scenario. We use the street canyon of [6] (Fig. 1): buildings 18.2 m apart, two 3.5-m lanes, and sidewalks on both sides; actors and UEs move on a periodic section of 80 m. Two O-RUs with 8×8 half-wavelength arrays at 28 GHz form two neighboring cells that share this section: O-RU 0 on the north side, either on a lamppost (5 m high) or on a facade (8 m), and O-RU 1 on a 5-m lamppost on the south sidewalk, 40 m further along the street. Pedestrian UEs at 1.5 m walk along the south sidewalk at 1.0–1.2 m/s. Each 60-s run has low or high traffic: 37.5 or 100 vehicles per lane and km (cars, buses, trucks at 6–12 m/s) and 6 or 12 pedestrians on the sidewalks plus 2 or 4 crossing the street. Paths are traced with Sionna RT [17], [18] (specular, order up to 2, at most 8 paths per link).

Service model. The SNR of a cell adds, over all traced paths, the path power with the full array gain after its model-B loss; this is an optimistic approximation of the received multipath power, since no single beam collects all paths with full gain. The UE is in outage when this total remaining power gives an SNR below $\gamma_{\mathrm{req}} = 9.3$ dB, the value needed for 400 Mbit/s; during 10 dB LoS events, the strongest surviving path of the same cell lies a median 26–33 dB below the unblocked LoS [6].

Handover model. The policy is the planner of [6]: every 0.1 s it receives the blocker tracks and the reported UE position, predicts the SNR of each cell by applying model B on the line between each O-RU and the reported position, and chooses the cell sequence that minimizes the predicted outage over a horizon of 0.5–3 s, with an interruption of 20 ms per handover; decisions take effect after a control-loop delay of 20 ms, and the sensing overhead of 2.3% of the resources is charged to it. The A5 thresholds and time-to-trigger are tuned per margin on the tuning seeds; the tuned time-to-trigger, 40 ms, is the smallest value of the grid. With perfect blocker tracks and a white Gaussian UE position error, it satisfies (2) only up to a per-axis standard deviation of about 8–13 cm, i.e., an RMS distance error of about 11–18 cm or a median distance error of about 9–15 cm [6].

Blocked LoS. An O-RU is blocked for a UE at an epoch when the model-B loss of its LoS path is at least 10 dB. In the biased model, the signal fed to the estimator replaces the blocked LoS by a detour around its dominant blocker (the

TABLE I

SINGLE-EPOCH ESTIMATOR ERROR AT FIXED GEOMETRIES (500 DRAWS EACH, MAIN HARDWARE, TDOA, NO EKF) VERSUS THE SINGLE-EPOCH LOS-ONLY PEB (BIASED MODEL FOR A BLOCKED LOS); ALL VALUES IN CM; RMSE^2 = BIAS^2 + CENTERED RMS^2 (EQ. (4)).

			400 MHz				100 MHz			
Mount	LoS state	UE x [m]	PEB	bias	centered	RMSE	PEB	bias	centered	RMSE
lamppost	both LoS	-20.0	1.5	1.3	1.7	2.2	1.5	7.7	7.4	10.7
lamppost	both LoS	20.0	4.2	1496.1	55.4	1497.1	4.5	875.6	345.0	941.2
lamppost	blocked (O-RU 0)	21.9	46.3	301.3	4.9	301.3	91.5	286.1	9.3	286.3
facade	both LoS	-20.0	3.3	31.9	4.8	32.2	3.4	30.0	16.1	34.1
facade	both LoS	20.0	6.7	88.6	6.1	88.8	7.1	94.0	9.4	94.5
facade	blocked (O-RU 0)	21.9	90.8	189.2	7.0	189.3	208.0	185.0	10.4	185.3

blocker with the largest model-B loss): with $\mathbf{q}$ the shorter of the detour points over the top and past either side of the blocker box, next to the point of the LoS closest to the blocker, the path keeps the attenuated complex gain of the LoS but has delay $(\|\mathbf{q}-\mathbf{p}_c\|+\|\mathbf{p}-\mathbf{q}\|)/c_0$ and arrival direction along $\mathbf{q}-\mathbf{p}_c$; reflected paths are only attenuated.

Signals and hardware. The UE transmits one SRS symbol per 0.1-s epoch at 23 dBm, spread uniformly over the active subcarriers; the O-RU noise figure is 7 dB. Numerology 3 (120 kHz) with 100, 200, and 400 MHz carriers uses 792, 1584, and 3168 active subcarriers [19]. We sweep σ_{sync} of 0, 1 and 3 ns (drawn once per run, a bias that the EKF cannot average out; 40 realizations over the held-out runs) and σ_ϕ of 0, 2 and 5 deg (fixed per O-RU and run). The main configuration uses TDoA, 400 MHz, $\sigma_{\mathrm{sync}}=1$ ns, $\sigma_\phi=2$ deg, and the biased blocked-LoS model.

Protocol. Estimator parameters were tuned on seeds 101–105 and methods fixed on development seeds 1001–1010. An earlier held-out set was consumed before the front/back disambiguation was added, so all results come from a second, never-used set of ten seeds (3001–3010), each simulated for two mounts and two traffic densities. The seed is the statistical unit: we average its runs, compare conditions with the exact two-sided Wilcoxon signed-rank test over the ten seeds, and give 95% bootstrap intervals over seeds. The map-aided tracker was developed after the dominant-path estimator, on the same tuning and development seeds, and was run once on the held-out seeds with unchanged parameters.

B. What Limits the Bound

Fig. 2 shows the median bounds versus bandwidth. With LoS to both O-RUs, the LoS-only PEB is below 10 cm in 84% of the epochs at 400 MHz and still in 81% at 100 MHz. Relaxing one limit at a time at 200 MHz scales the median LoS-only PEB (4.8 cm) by 0.97 for twice the bandwidth, 0.94 for ten times the SNR, 0.72 for perfect synchronization, and 0.07 for perfect calibration. On these short links, the delay and angle measurements are precise enough that the element phase errors, which bias the angles, dominate. Without hardware errors, the LoS-only PEB is below 10 cm in 92% of the epochs; with 3 ns and 5 deg errors, in 44%. The map-aided PEB is far less sensitive (99% and 98%), because reflections from known surfaces add independent geometric information.

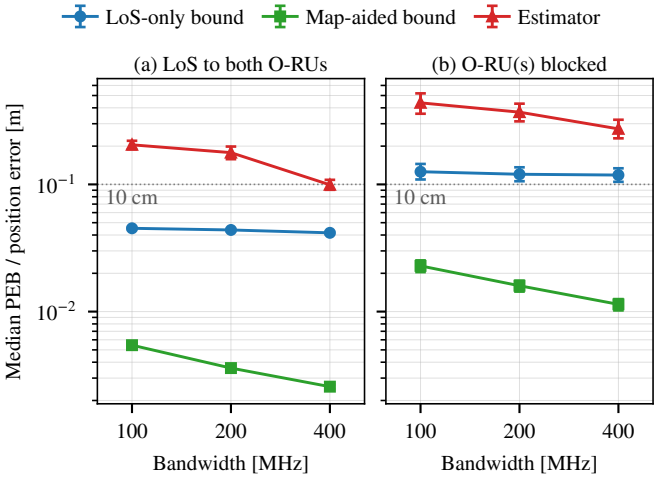


Fig. 2. Median single-epoch position error bound (LoS-only, map-aided) and median estimator error versus bandwidth, for epochs (a) with LoS to both O-RUs and (b) with at least one blocked O-RU; main configuration (95% CI). The bound and the error are different quantities (Section IV). The LoS-only bound barely changes with bandwidth because array calibration limits it (Section V-B).

C. Blockage and the Digital Twin

In the biased model, the median LoS-only PEB rises by a factor of 2.8 during blockage ($p=0.002$), and only 47% of the blocked epochs remain below 10 cm. If the attenuated LoS kept its geometry, the factor would be only 1.2, which shows how optimistic it is to apply model B's power loss without its geometric consequence. Map-aided information lowers the PEB during blockage to 0.09 times the LoS-only value ($p=0.002$). Along the street, the LoS-only PEB varies with the geometry of the two O-RUs from about a centimeter to over a decimeter, while the map-aided PEB stays mostly in the millimeter range. A low map-aided PEB indicates that the information is present in the signal; whether an estimator can extract it is a separate question.

D. Practical Estimators and the Bound

At 400 MHz, the dominant-path estimator reaches a median error of 12 cm (10 cm with LoS to both O-RUs, 27 cm during blockage); 45% of its errors are below 10 cm, and its 90th percentile is 2.10 m. At 100 MHz, the median grows to 24 cm.

TABLE II
ESTIMATOR ERROR AROUND REFERENCE-CELL BLOCKAGE EVENTS (10 DB, MAIN CONFIGURATION).

Window	Median	90th perc.	>0.1 m	>0.25 m
1.0–0.5 s before onset	12 cm	4.24 m	54%	30%
0.5–0 s before onset	11 cm	4.49 m	53%	29%
During the event	87 cm	3.26 m	88%	72%
Elsewhere	10 cm	1.59 m	49%	29%

To compare the estimator with the bound on equal terms, we ran 500 independent noise and hardware draws through the single-epoch estimator (without EKF) at 6 fixed geometries and decomposed its error as

$$\mathrm{RMSE}^2 = \|\mathbb{E}[\hat{\mathbf{p}}] - \mathbf{p}\|^2 + \mathbb{E}\big[\|\hat{\mathbf{p}} - \mathbb{E}[\hat{\mathbf{p}}]\|^2\big], \qquad (4)$$

i.e., squared bias plus squared centered RMS (Table I). The hardware errors are redrawn in every draw, so their contribution appears in the centered RMS, whereas the bias contains only deterministic errors such as unresolved multipath and the detoured LoS; within a run, a hardware realization is fixed and acts as a bias. The squared bias accounts for most of the mean squared error in five of the six geometries at 400 MHz (98–100% of it); in the remaining one (lamppost, both LoS, $x = -20$ m), the squared bias is 38% of it and the random spread is larger. The bias is caused by interference of unresolved multipath, and in one geometry by a front/back mirror solution. The centered RMS is geometry dependent, between 0.077–13 times the PEB at 400 MHz: close to the bound at some geometries and an order of magnitude above it at others. In blocked geometries it falls below the PEB, which is possible because the estimator is biased: it uses the detoured LoS precisely but in the wrong place. Over the held-out runs, front/back mirror solutions remain in 5% of the epochs and dominate the squared error (RMS 6.36 m), while reflected dominant paths set the 90th percentile. A variant that used reflections as virtual anchors did not reduce the tail on the development seeds (90th percentile 3.52 vs. 1.87 m) and was not pursued.

The map-aided tracker reaches a median error of 3.7 cm (4.5 cm with at least one O-RU blocked) and a 90th percentile of 0.25 m, significantly better than the dominant-path estimator ($p = 0.002$ for all three statistics). Its largest errors occur in the 1 s after a UE re-enters the periodic street section, a simulation artifact in which the track keeps the pre-jump position until it re-initializes; these windows contain 1.7% of the epochs and are not excluded.

E. Positioning Errors Around Blockage Onsets

Table II shows the estimator error around the 752 blockage events of the reference cell, defined per UE as the cell with the larger unblocked received power; an event is an interval with LoS loss of at least 10 dB, with gaps below 0.5 s merged; each epoch is counted in one window, with priority to the event, and the first 2 s of each run are excluded. In the second before onset, a window relevant to proactive handover, the median error and the share of errors above 0.25 m are about as elsewhere, but the large errors become larger: the 90th percentile rises from 1.59 to 4.49 m. During the event, most of the distribution deteriorates, with the median growing to 87 cm. Errors thus worsen in the time windows relevant to proactive handover: before a blockage, the extreme errors grow; during it, the bulk of the errors does, which the overall median hides. For the map-aided tracker, the 90th percentile does not rise before onset (0.20 vs. 0.35 m elsewhere), and its median during the event is 6.4 cm. The windows refer to the blockage events of a reference cell, not to the decision times of a particular planner.

TABLE III
PLANNER OUTAGE MINUS A5 [S/UE-MIN] BY SOURCE OF THE UE POSITION, WITH PERFECT BLOCKER TRACKS. NEGATIVE VALUES (SHADED) MEAN LESS OUTAGE THAN A5; †: NOT SIGNIFICANT (SEED-LEVEL WILCOXON $p \geq 0.05$; POINTWISE TESTS, EXPLORATORY). REF. = 3GPP REFERENCE BUDGET (35.9 DB MARGIN). *ORACLE DIAGNOSTIC: MIRROR SOLUTIONS REPLACED BY THE SINGLE-EPOCH HYPOTHESIS CLOSEST TO THE TRUE POSITION.

UE position	Link margin				Ref.
	15 dB	20 dB	25 dB	30 dB	
Reference inputs					
exact	−0.108	−0.066†	−0.061	−0.007†	−0.025
white 1 m error	+1.402	+0.824	+0.476	+0.290	+0.117
Synthetic, bound level (full covariance)					
map-aided	−0.099	−0.049†	−0.044	+0.021†	−0.005†
LoS-only	+0.162	+0.301	+0.175	+0.151	+0.097
LoS-only, no fix held	+0.071†	+0.142†	+0.058†	+0.073†	+0.036†
Estimated					
map-aided tracker	+0.009†	+0.031†	+0.072†	+0.015†	−0.002†
dominant-path	+0.652	+0.507	+0.284	+0.118	+0.052
ideal hardware	+0.584	+0.483	+0.295	+0.121	+0.054
capped at p90	+0.525	+0.434	+0.251	+0.118	+0.051
mirror-corr.*	+0.461	+0.443	+0.260	+0.115	+0.048

F. Implications for Handover

Table III feeds positions into the planner, with perfect blocker tracks so that the user position is the only source of error. With exact positions, the planner has a lower outage than A5 from 15 dB margin upward (significantly at 15 and 25 dB and at the reference). As a sensitivity test, we feed synthetic errors $\mathbf{e}_t \sim \mathcal{N}\big(\mathbf{0}, \mathbf{J}_{\mathbf{p}}(t)^{-1}\big)$, drawn independently per epoch from the single-epoch bound of the respective model, which is anisotropic (median ratio of principal standard deviations 3.0). Errors at the map-aided bound preserve a lower mean outage than A5 at most tested margins, with significant improvements at 15 and 25 dB; errors at the LoS-only bound, which grow during blockage, make it worse than A5 at every margin. With the positions of the dominant-path estimator, the planner is significantly worse than A5 at every margin, also with ideal hardware; with those of the map-aided tracker, it is not significantly different from A5 at any margin ($p \geq 0.084$), although at 25 dB its bootstrap interval lies above zero. Capping each error of the EKF output at the pooled 90th percentile (1.90 m) or replacing its mirror solutions by the single-epoch hypothesis closest to the true

position (an oracle diagnostic, not refiltered) reduces the loss but leaves it significant, so the largest errors are not its only cause. Synthetic bound-level errors are an optimistic model; they show what positioning would have to achieve, not that a practical estimator achieves it. In the 3.0% of epochs in which both O-RUs are blocked, the LoS-only bound is singular; these epochs are treated as having no fix ($100\,\mathrm{m}$ standard deviation). Holding the last valid position instead keeps the LoS-only row above A5 at every margin but removes its significance, so that row depends mainly on how missing fixes are handled. The synthetic errors are independent across epochs, whereas hardware errors persist over a run; the effect of temporal correlation on the planner is not evaluated here. The absolute gains at stake are small: at $15\,\mathrm{dB}$ margin, A5 has an outage of 1.88 s/UE-min, and the relative change is -5.7% with exact positions and -5.2% with bound-level map-aided positions; at the reference, A5 has 0.058 s/UE-min, and the changes are -43.1% and $-8.2\%^{\dagger}$, respectively; the application-level impact is not evaluated here.

VI. Discussion and Limitations

For blockage-aware handover, the relevant accuracy is the error around blockage onsets, not the overall median. The gap between the bound and the dominant-path estimator is mainly systematic multipath error, and a simple virtual-anchor variant did not close it; the map-aided tracker, which resolves several paths and associates them probabilistically with known surfaces, does extract much of the information that the map-aided bound indicates. With its positions, proactive handover is no longer significantly worse than A5, but it does not reach the small gain that exact positions provide; whether this remaining gap stems from the tracker's residual errors, including those after the UE jumps, is not resolved here. The study has limitations: a single street canyon with two O-RUs; a tracker added after the first evaluation and run with an exact map; specular paths of order at most 2 and no diffuse scattering; a perfectly known map in the map-aided bound; the biased model of a blocked LoS as our modeling choice; bodies as opaque knife-edge screens, without window transmission, vehicle-body reflections, or self-blockage by the user; single-epoch bounds without the temporal information of tracking; and planar arrays of isotropic elements, whose front/back ambiguity a physical back plane would remove in hardware [8].

VII. Conclusion

Under the tested configurations, the single-epoch bound is limited by array calibration rather than bandwidth; blockage substantially degrades the LoS-only bound, while known reflected paths preserve a much lower map-aided bound. A dominant-path estimator shows substantial systematic multipath errors that worsen around blockage, and its positions make proactive handover worse than reactive handover. A map-aided multipath tracker reduces the median error to a few centimeters and keeps it low around blockage onsets, so that proactive handover is no longer significantly worse than the 3GPP event; reaching the small gain that exact positions provide remains open.

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